

ZHUFAN GU

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EDUCATION

Trine University, Phoenix, AZ

Master of Science in Information Studies

May 2025 – May 2027

- Courses: Database, Object-oriented Programming, Cybersecurity, Software Testing & Quality Assurance

Santa Clara University, Santa Clara, CA

Master of Science in Computer Science and Engineering

Sep 2022 – Dec 2025

- Courses: Data Structures & Algorithms, Distributed Systems, Operating Systems, Computer Networks

University of St. Andrews, St. Andrews, UK

Master of Science in Human Computer Interaction

Sep 2019 – Dec 2020

EXPERIENCE

CodeDay Labs | *Software Engineer Intern*

Jun 2024 – Aug 2024

- Developed a test framework in **Rust** to validate the correctness of algorithms in an algebraic simplification solver.
- Exposed the Rust-based solver as a high-performance **Python** library using **PyO3** and **Maturin**.

Open Energy Dashboard | *Software Engineer Intern*

Jun 2023 – Aug 2023

- Built a unit conversion engine in **JavaScript** to enable dual-unit reporting for federal compliance standards.
- Validated numerical accuracy against CSV datasets within strict precision thresholds using **Mocha.js**.
- Standardized the development environment with **Docker**, reducing setup time and ensuring consistent builds.

OPEN SOURCE EXPERIENCE

React-PDF (10.7k stars)

- Optimized accessibility of PDF links to reach a 96/100 Lighthouse score and WCAG compliance by generating ARIA labels parsed from PDF annotation metadata.

PROJECTS

Pantry AI | *Full-Stack Web Development*

- Architected a multimodal AI application using **Next.js/FastAPI**, featuring ingredient recognition and recipe generation.
- Deployed a serverless backend on **AWS Lambda**, utilizing **Docker** to ensure binary compatibility between local ARM development and the x86 cloud runtime.

AppliTrack | *Full-Stack Web Development*

- Engineered a **MERN** stack job tracking platform, securing APIs with **JWT** authentication, Role-Based Access Control (**RBAC**), and validation middleware.
- Optimized frontend data synchronization and caching using **TanStack Query**, reducing redundant network requests.

SKILLS

Languages: Python, Rust, C++, Java, JavaScript/TypeScript, HTML

Frameworks & Libraries: FastAPI, Node.js, Express.js, Next.js, React, PyO3, Maturin

Databases: PostgreSQL, MongoDB, Oracle

Cloud & DevOps: AWS (Lambda, API Gateway, Amplify), Docker, GitHub Actions, Render, Git

Testing: Pytest, Jest, Playwright, Postman

HONORS AND REWARDS

- Zhejiang Government Scholarship (Top 3%)
- Excellent Graduates of Zhejiang Province